EXP:3 – ASSIGNMENT PROBLEM

AIM:

To understand and solve the assignment problem using python ,assignment team of workers.

Procedure:

1. Input Data: Define the cost matrix where each element represents the cost of assigning a particular task to a specific worker.

2. Algorithm: Use an optimization algorithm like the Hungarian algorithm (or the Munkres algorithm) to find the optimal assignment that minimizes the total cost.

3. Output: Display the optimal assignment along with the minimum total cost.

CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Define cost matrix
cost_matrix = np.array([
    [9, 2, 7, 8],
    [6, 4, 3, 7],
    [5, 8, 1, 8],
    [7, 6, 9, 4]
])

# Solve assignment problem
row_indices, col_indices = linear_sum_assignment(cost_matrix)

# Calculate total cost
total_cost = cost_matrix[row_indices, col_indices].sum()

# Print results
print("Optimal Assignment:")
```

```
for i, j in zip(row_indices, col_indices):
    print(f"Task {i+1} assigned to Worker {j+1}")

print("\nMinimum Total Cost:", total_cost)
```

Output:

Optimal Assignment:

Task 1 assigned to Worker 2

Task 2 assigned to Worker 1

Task 3 assigned to Worker 3

Task 4 assigned to Worker 4

Minimum Total Cost: 13

Result:

The given assignment problem was solved successfully using Python. The optimal allocation of tasks to workers was determined using the Hungarian algorithm, and the minimum total cost obtained is ₹13.